

# E-commerce

## Building an E-commerce Site

*Sixth Edition*

# Building an E-commerce Site: A Systematic Approach

- Most important management challenges:
  - Developing a clear understanding of business objectives
  - Knowing how to choose the right technology to achieve those objectives

# Pieces of the Site-Building Puzzle

- Main areas where you will need to make decisions:
  - Human resources and organizational capabilities
    - Creating team with skill set needed to build and manage a successful site
  - Hardware
  - Software
  - Telecommunications
  - Site design

# The Systems Development Life Cycle

- Methodology for understanding business objectives of a system and designing an appropriate solution
- Five major steps:
  1. Systems analysis/planning
  2. Systems design
  3. Building the system
  4. Testing
  5. Maintenance and evolution

# Web Site Systems Development Life Cycle

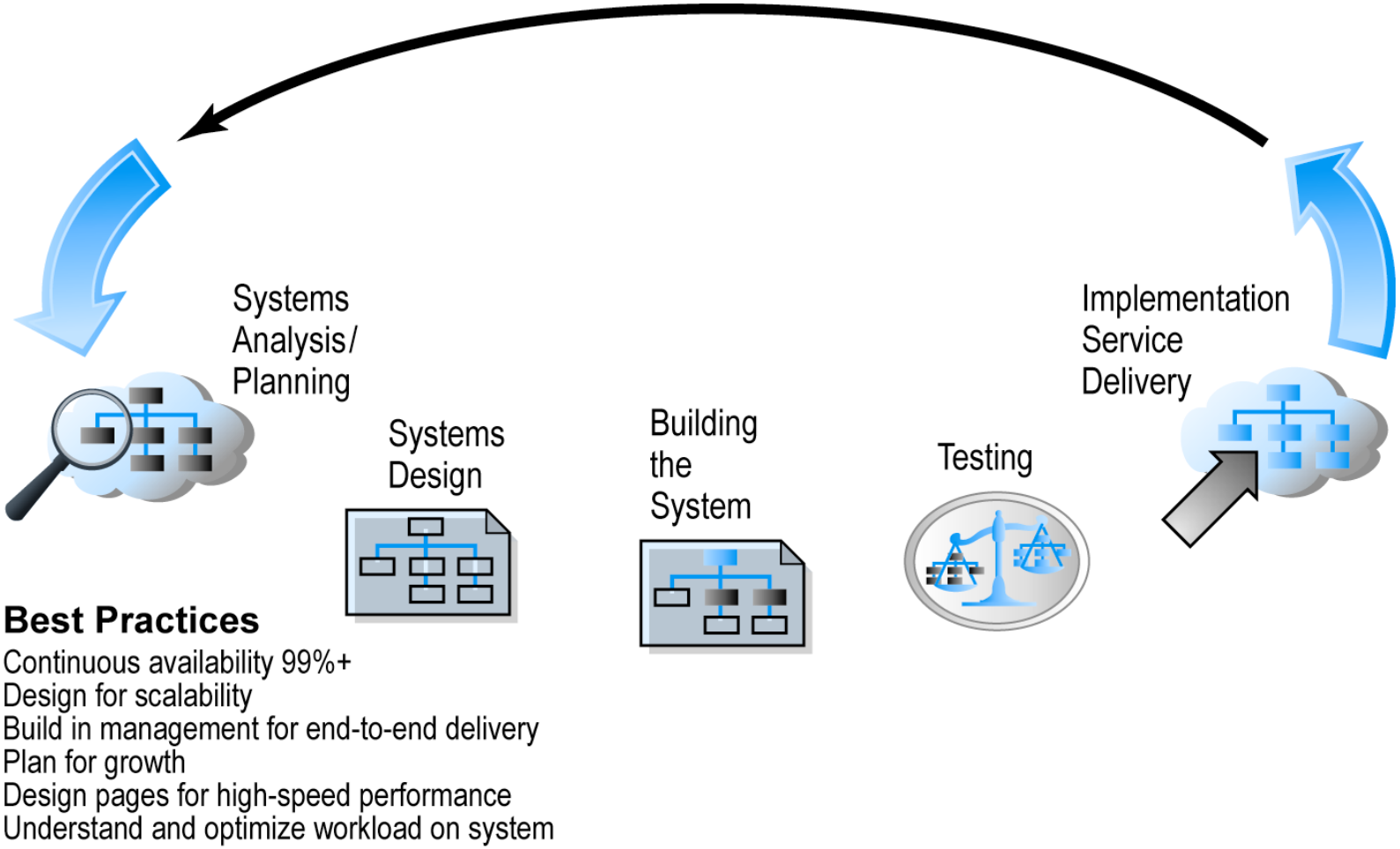



**Figure 4.2, Page 208**

# System Analysis/Planning

- Business objectives:
  - List of capabilities you want your site to have
- System functionalities:
  - List of information system capabilities needed to achieve business objectives
- Information requirements:
  - Information elements that system must produce in order to achieve business objectives

**TABLE 4.1** SYSTEM ANALYSIS: BUSINESS OBJECTIVES, SYSTEM FUNCTIONALITY, AND INFORMATION REQUIREMENTS FOR A TYPICAL E-COMMERCE SITE

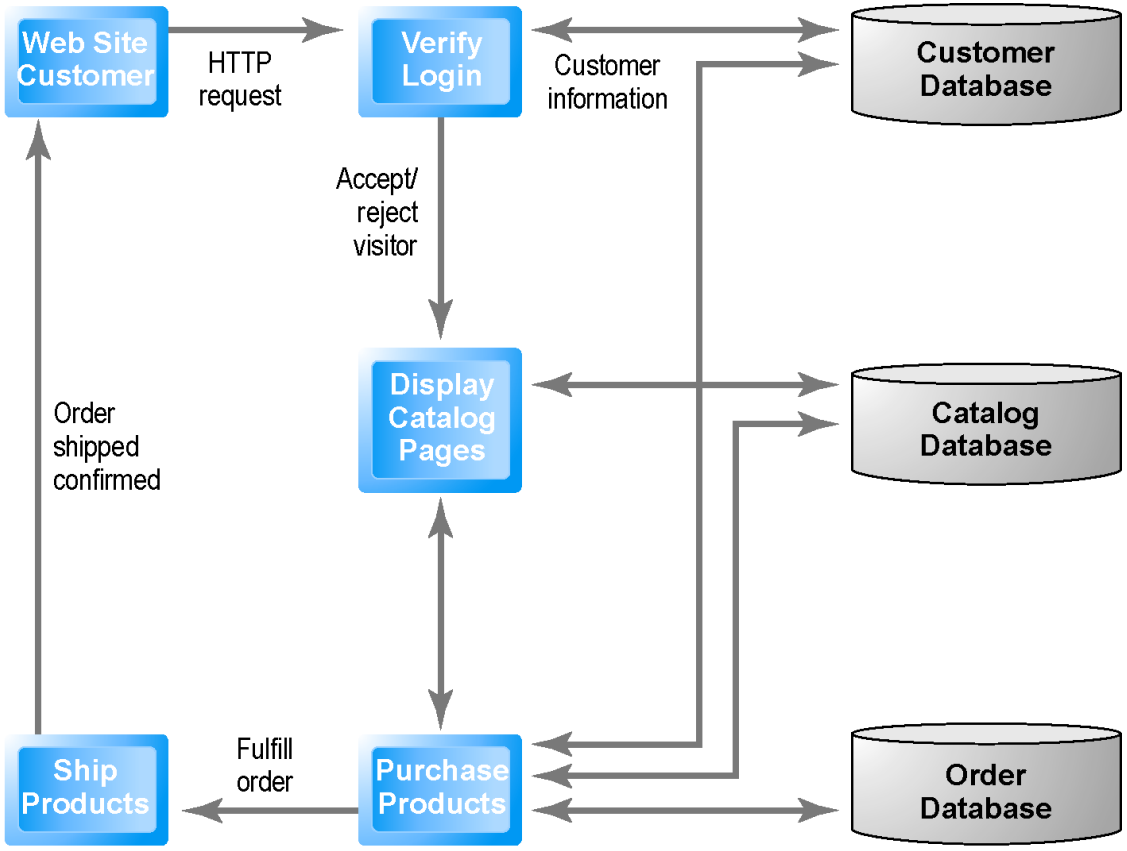
| BUSINESS OBJECTIVE | SYSTEM FUNCTIONALITY | INFORMATION REQUIREMENTS |
|---|---|---|
| Display goods | Digital catalog | Dynamic text and graphics catalog |
| Provide product information (content) | Product database | Product description, stocking numbers, inventory levels |
| Personalize/customize product | Customer on-site tracking | Site log for every customer visit; data mining capability to identify common customer paths and appropriate responses |
| Execute a transaction payment | Shopping cart/payment system | Secure credit card clearing; multiple options |
| Accumulate customer information | Customer database | Name, address, phone, and e-mail for all customers; online customer registration |
| Provide after-sale customer support | Sales database | Customer ID, product, date, payment, shipment date |
| Coordinate marketing/advertising | Ad server, e-mail server, e-mail, campaign manager, ad banner manager | Site behavior log of prospects and customers linked to e-mail and banner ad campaigns |
| Understand marketing effectiveness | Site tracking and reporting system | Number of unique visitors, pages visited, products purchased, identified by marketing campaign |
| Provide production and supplier links | Inventory management system | Product and inventory levels, supplier ID and contact, order quantity data by product |

**Table 4.1, Page 209**

# Systems Design:
## Hardware and Software Platforms

- System design specification:
  - Description of main components of a system and their relationship to one another
- Two components of system design:
  1. Logical design
     - Data flow diagrams, processing functions, databases
  2. Physical design
     - Specifies actual physical, software components, models, etc.

# Logical Design for a Simple Web Site

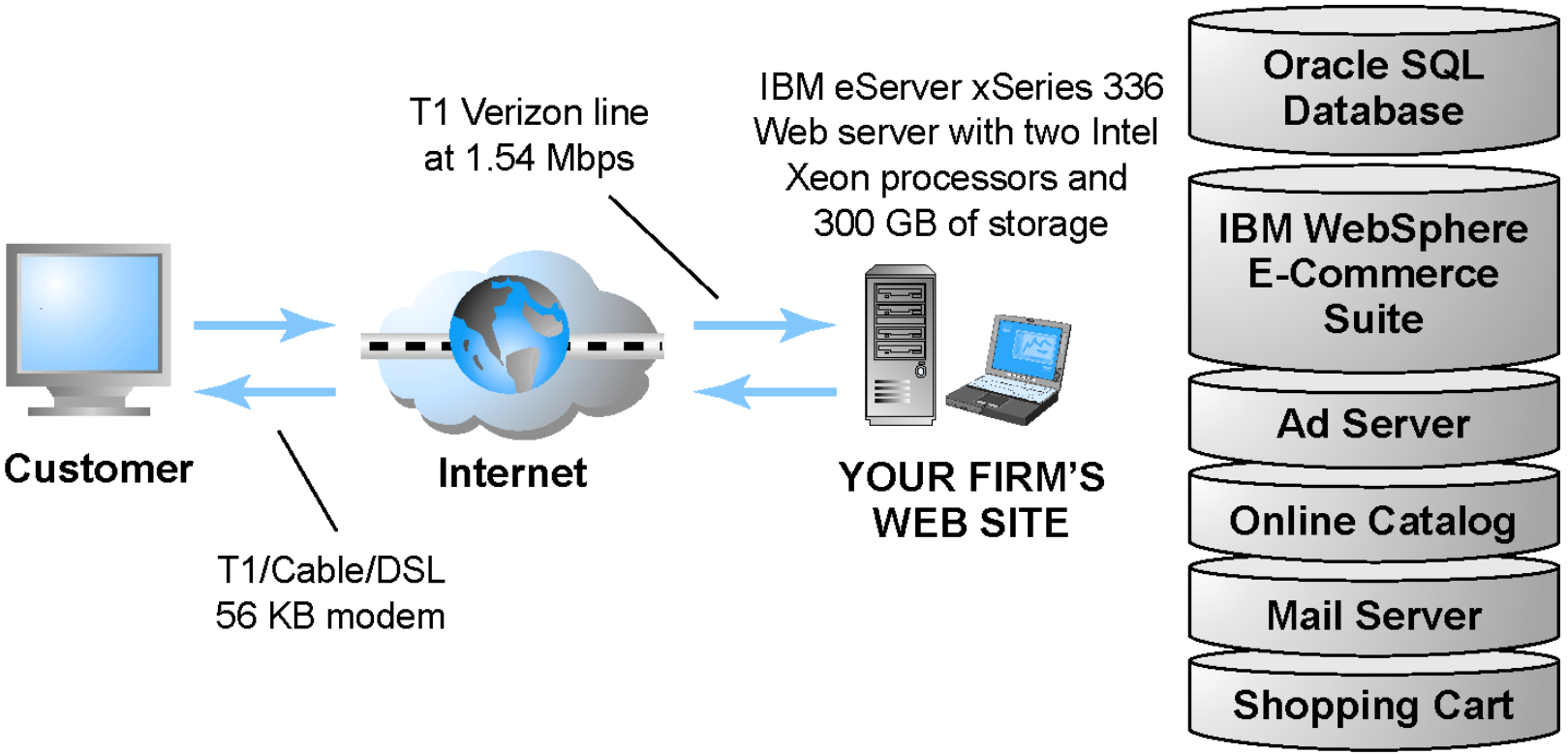



**(a) Simple Data Flow Diagram**

This data flow diagram describes the flow of information requests and responses for a sample Web site

**Figure 4.3 (a), Page 211**

# Physical Design for a Simple Web Site




**(b) Simple Physical Design**

A physical design describes the hardware and software needed to realize the logical design

**Figure 4.3 (b), Page 211**

# Build/Host Your Own versus Outsourcing

- Outsourcing: hiring vendors to provide services involved in building site
- Build own vs. outsourcing:
  - Build your own requires team with diverse skill set; choice of software tools; both risks and possible benefits
- Host own vs. outsourcing
  - Hosting: hosting company responsible for ensuring site is accessible 24/7, for monthly fee
  - Co-location: firm purchases or leases Web server (with control over its operation), but server is located at vendor's facility

# Choices in Building and Hosting

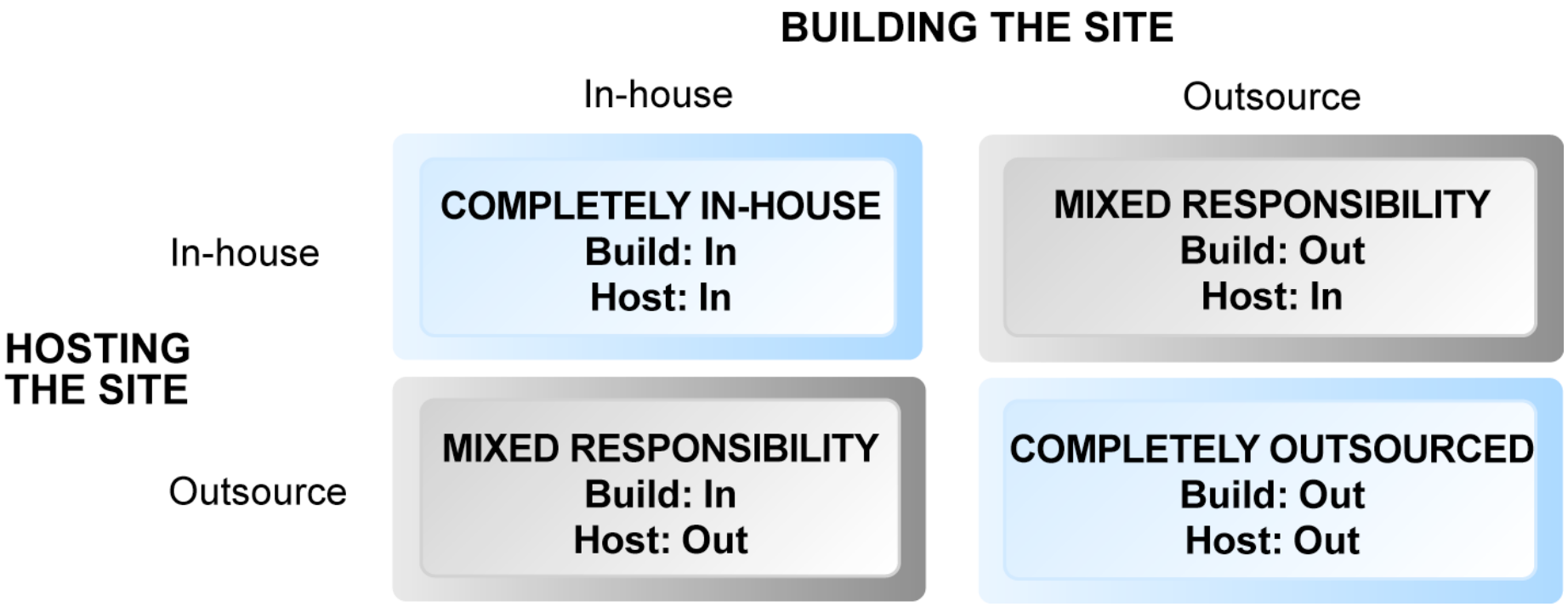



Figure 4.4, Page 212

# Testing, Implementation, and Maintenance

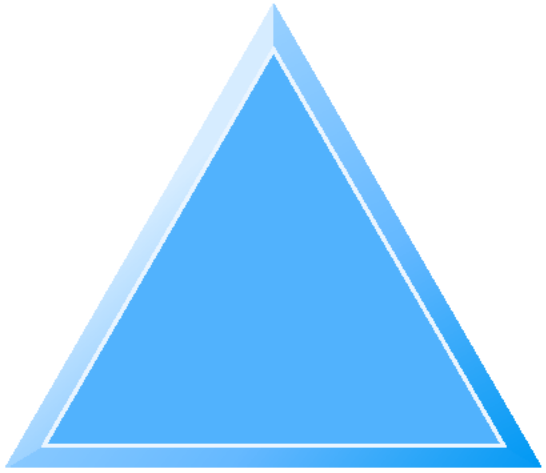
- Testing
  - Unit testing
  - System testing
  - Acceptance testing
- Implementation and maintenance:
  - Maintenance is ongoing
  - Maintenance costs: parallel to development costs
  - Benchmarking:

    is the practice of comparing business processes and performance metrics to industry bests and best practices from other companies

# Factors in Web Site Optimization

**Page Delivery**

Content delivery networks
Edge caching
Bandwidth



**Page Generation**

Server response time
Device-based accelerators
Efficient resource allocation
Resource utilization thresholds
Monitoring site performance

**Page Content**

Optimize HTML
Optimize images
Site architecture
Efficient page style

**Figure 4.7, Page 219**

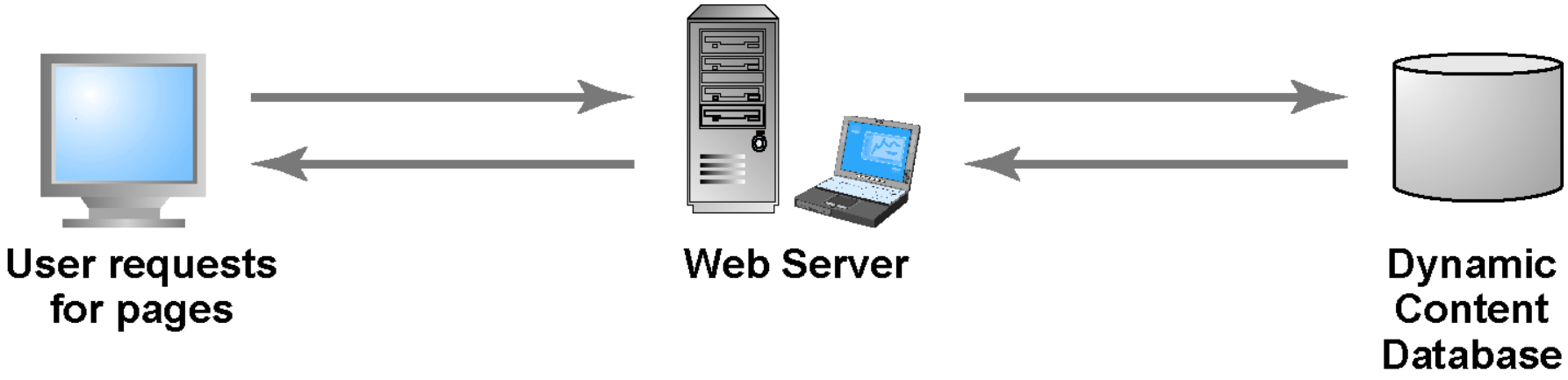
# Web Site Budgets

- From $5,000 to millions of dollars/year
- Components of budget:
  - System maintenance
  - System development
  - Content design & development
  - Hardware
  - Telecommunications
  - Software

# Simple versus Multi-tiered Web Site Architecture

- System architecture
  - Arrangement of software, machinery, and tasks in an information system needed to achieve a specific functionality
- Two-tier
  - Web server and database server
- Multi-tier
  - Web application servers
  - Backend, legacy databases

# Two-Tier E-commerce Architecture




**(a) Two-tier Architecture**

In a two-tier architecture, a Web server responds to requests for Web pages and a database server provides backend data storage.

Figure 4.9(a), Page 221

# Multi-tier E-commerce Architecture




**(b) Multi-tier Architecture**

In a multi-tier architecture, a Web server is linked to a middle-tier layer that typically includes a series of application servers that perform specific tasks, as well as to a backend layer of existing corporate systems.

**Figure 4.9(b), Page 221**

# Web Server Software

- Apache
  - Leading Web server software (47% of market)
  - Works only with UNIX, Linux OSs
- Microsoft's Internet Information Server (IIS)
  - Second major Web server software (25% of market)
  - Windows-based

| TABLE 4.3 | BASIC FUNCTIONALITY PROVIDED BY WEB SERVERS |
|---|---|
| FUNCTIONALITY | DESCRIPTION |
| Processing of HTTP requests | Receive and respond to client requests for HTML pages |
| Security services (Secure Sockets Layer) | Verify username and password; process certificates and private/public key information required for credit card processing and other secure information |
| File Transfer Protocol | Permits transfer of very large files from server to server |
| Search engine | Indexing of site content; keyword search capability |
| Data capture | Log file of all visits, time, duration, and referral source |
| E-mail | Ability to send, receive, and store e-mail messages |
| Site management tools | Calculate and display key site statistics, such as unique visitors, page requests, and origin of requests; check links on pages |

**Table 4.3, Page 223**

# Site Management Tools

- Basic tools
  - Included in all Web servers
  - Verify that links on pages are still valid
  - Identify orphan files
- Third-party software and services for advanced site management
  - Monitor customer purchases, marketing campaign effectiveness, etc.
  - E.g. WebTrends Analytics 9, Google Analytics

# Dynamic Page Generation Tools

- Dynamic page generation:
  - Contents of Web page stored as objects in database and fetched when needed
- Common tools: CGI, ASP, JSP
- Advantages
  - Lowers menu costs
  - Permits easy online market segmentation
  - Enables cost-free price discrimination
  - Enables Web content management system (WCMS)

# Application Servers

- Web application servers:
  - Provide specific business functionality required for a Web site
  - Type of middleware
    - Isolate business applications from Web servers and databases
  - Single-function applications increasingly being replaced by integrated software tools that combine all functionality needed for e-commerce site

**TABLE 4.4 APPLICATION SERVERS AND THEIR FUNCTION**

| APPLICATION SERVER | FUNCTIONALITY |
|---|---|
| Catalog display | Provides a database for product descriptions and prices |
| Transaction processing (shopping cart) | Accepts orders and clears payments |
| List server | Creates and serves mailing lists and manages e-mail marketing campaigns |
| Proxy server | Monitors and controls access to main Web server; implements firewall protection |
| Mail server | Manages Internet e-mail |
| Audio/video server | Stores and delivers streaming media content |
| Chat server | Creates an environment for online real-time text and audio interactions with customers |
| News server | Provides connectivity and displays Internet news feeds |
| Fax server | Provides fax reception and sending using a Web server |
| Groupware server | Creates workgroup environments for online collaboration |
| Database server | Stores customer, product, and price information |
| Ad server | Maintains Web-enabled database of advertising banners that permits customized and personalized display of advertisements based on consumer behavior and characteristics |
| Auction server | Provides a transaction environment for conducting online auctions |
| B2B server | Implements buy, sell, and link marketplaces for commercial transactions |

**Table 4.4, Page 227**

# E-commerce Merchant Server Software

- Provides basic functionality for online sales
  - Online catalog
    - List of products available on Web site
  - Shopping cart
    - Allows shoppers to set aside, review, edit selections, and then make purchase
  - Credit card processing
    - Typically works in conjunction with shopping cart
    - Verifies card and puts through credit to company's account at checkout

# Merchant Server Software Packages

- Integrated environment with most or all of functionality needed
- Key factors in selecting a package
  - Functionality
  - Support for different business models
  - Business process modeling tools
  - Visual site management and reporting
  - Performance and scalability
  - Connectivity to existing business systems
  - Compliance with standards
  - Global and multicultural capability
  - Local sales tax and shipping rules

# Building Your Own E-commerce Site

- Options for small firms
  - Hosted e-commerce sites, e.g., Yahoo's Merchant Solutions
    - Site building tools
    - E-commerce templates
  - Open-source merchant server software
    - Enables you to build truly custom site
    - Requires programmer with expertise, time

# Choosing the Hardware for an E-commerce Site

- Hardware platform:
  - Underlying computing equipment that system uses to achieve e-commerce functionality
- Objective:
  - Enough platform capacity to meet peak demand without wasting money
- Important to understand the different factors that affect speed, capacity, and scalability of a site

# Right-Sizing Your Hardware Platform: The Demand Side

- Demand is the most important factor affecting speed of site
- Factors in overall demand:
  - Number of simultaneous users in peak periods
  - Nature of customer requests (user profile)
  - Type of content (dynamic versus static Web pages)
  - Required security
  - Number of items in inventory
  - Number of page requests
  - Speed of legacy applications

**TABLE 4.7 FACTORS IN RIGHT-SIZING AN E-COMMERCE PLATFORM**

| SITE TYPE | PUBLISH/ SUBSCRIBE | SHOPPING | CUSTOMER SELF-SERVICE | TRADING | WEB SERVICES/B2B |
|---|---|---|---|---|---|
| Examples | WSJ.com | Amazon | Travelocity | E*Trade | Ariba e-procurement exchanges |
| Content | Dynamic<br>Multiple authors<br>High volume<br>Not user specific | Catalog<br>Dynamic items<br>User profiles with data mining | Data in legacy applications<br>Multiple data sources | Time sensitive<br>High volatility<br>Multiple suppliers and consumers<br>Complex transactions | Data in legacy applications<br>Multiple data sources<br>Complex transactions |
| Security | Low | Privacy<br>Non-repudiation<br>Integrity<br>Authentication<br>Regulations | Privacy<br>Non-repudiation<br>Integrity<br>Authentication<br>Regulations | Privacy<br>Non-repudiation<br>Integrity<br>Authentication<br>Regulations | Privacy<br>Non-repudiation<br>Integrity<br>Authentication<br>Regulations |
| Percent secure pages | Low | Medium | Medium | High | Medium |
| Cross session information | No | High | High | High | High |
| Searches | Dynamic<br>Low volume | Dynamic<br>High volume | Non dynamic<br>Low volume | Non dynamic<br>Low volume | Non dynamic<br>Moderate volume |
| Unique items (SKUs) | High | Medium to high | Medium | High | Medium to high |
| Transaction volume | Moderate | Moderate to high | Moderate | High to extremely high | Moderate |
| Legacy integration complexity | Low | Medium | High | High | High |
| Page views (hits) | High to very high | Moderate to high | Moderate to low | Moderate to high | Moderate |

**Table 4.7, Page 232**

# Right-Sizing Your Hardware Platform: The Supply Side

- Scalability:
  - Ability of site to increase in size as demand warrants
- Ways to scale hardware:
  - Vertically
    - Increase processing power of individual components
  - Horizontally
    - Employ multiple computers to share workload
  - Improve processing architecture

**TABLE 4.8 VERTICAL AND HORIZONTAL SCALING TECHNIQUES**

| TECHNIQUE | APPLICATION |
|---|---|
| Use a faster computer | Applies to edge servers, presentation servers, data servers, etc. |
| Create a cluster of computers | Use computers in parallel to balance loads |
| Use appliance servers | Special-purpose computers optimized for their task |
| Segment workload | Segment incoming work to specialized computers |
| Batch requests | Combine related requests for data into groups, process as a group |
| Manage connections | Reduce connections between processes and computers to a minimum |
| Aggregate user data | Aggregate user data from legacy applications in single data pools |
| Cache | Store frequently used data in cache rather than on the disk |

**Table 4.8, Page 236**

# Vertically Scaling a System

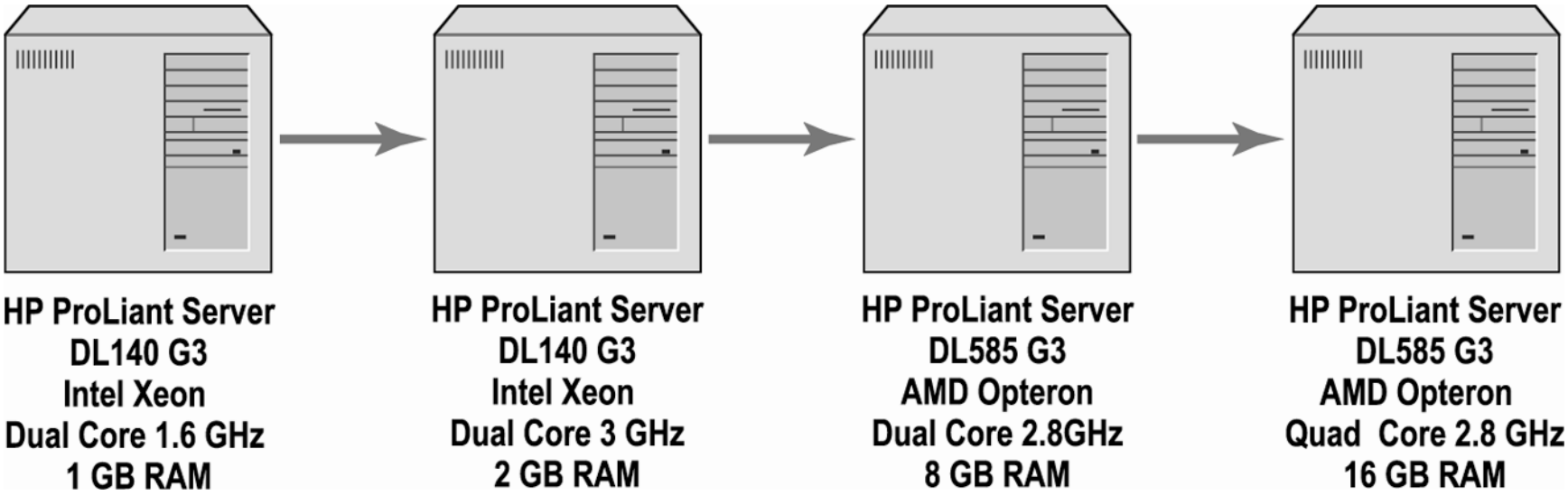



Figure 4.14, Page 237

# Horizontally Scaling a System

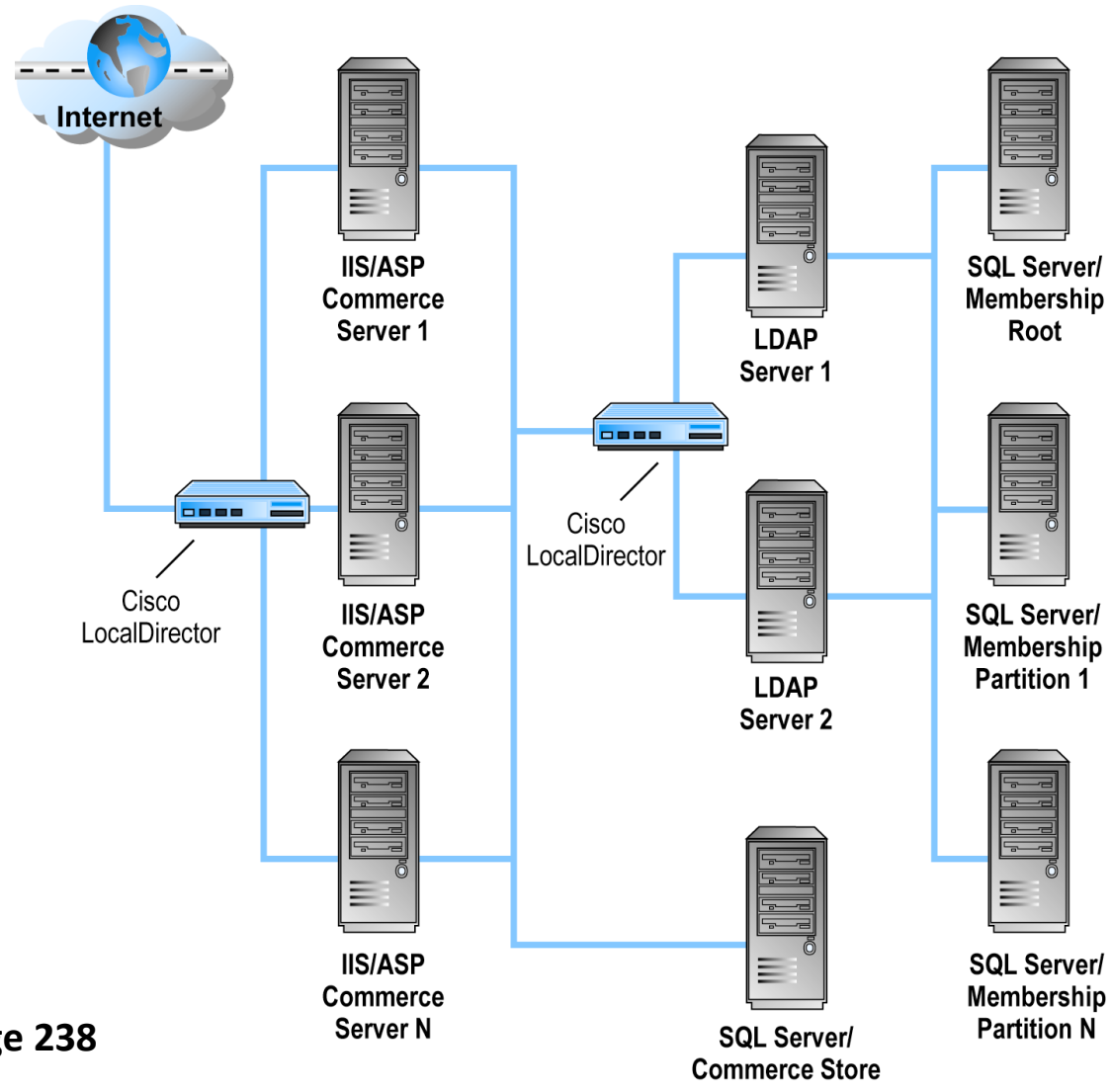



Figure 4.15, Page 238

## TABLE 4.9 IMPROVING THE PROCESSING ARCHITECTURE OF YOUR SITE

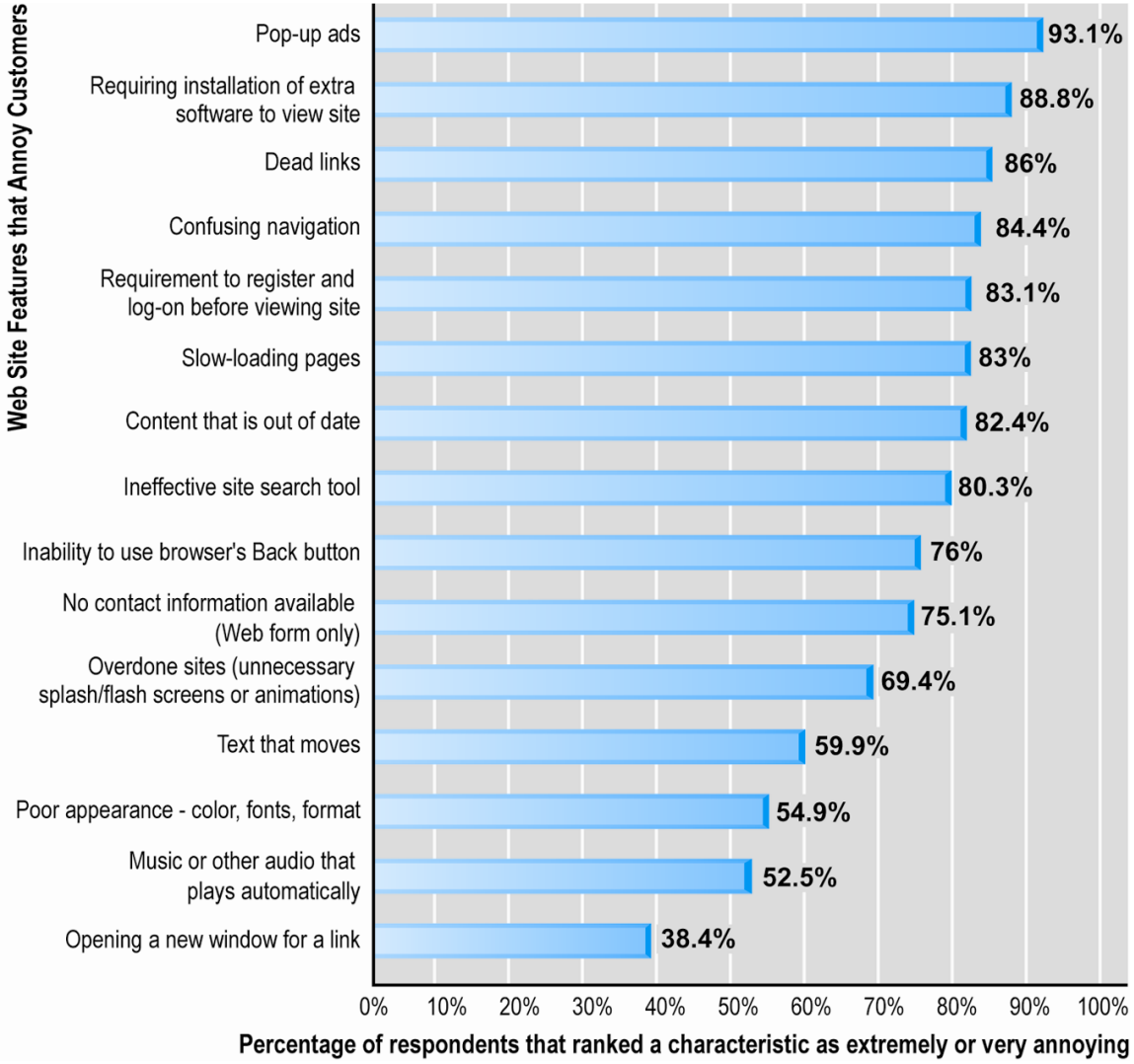
| ARCHITECTURE IMPROVEMENT | DESCRIPTION |
|---|---|
| Separate static content from dynamic content | Use specialized servers for each type of workload. |
| Cache static content | Increase RAM to the gigabyte range and store static content in RAM. |
| Cache database lookup tables | Cache tables used to look up database records. |
| Consolidate business logic on dedicated servers | Put shopping cart, credit card processing, and other CPU-intensive activity on dedicated servers. |
| Optimize ASP code | Examine your code to ensure it is operating efficiently. |
| Optimize the database schema | Examine your database search times and take steps to reduce access times. |

**Table 4.9, Page 239**

# Other E-Commerce Site Tools

- Web site design: Basic business considerations
  - Enabling customers to find and buy what they need
- Tools for Web site optimization
  - Search engine placement
    - Keywords, page titles
    - Identify market niches, localize site
    - Expertise
    - Links
    - Search engine ads

# E-commerce Web Site Features that Annoy Customers




**SOURCE: Based on data from Hostway Corporation's survey, Consumers' Pet Peeves about Commercial Web Sites, Hostway Corporation, 2007.**

**Figure 4.16, Page 240**

## TABLE 4.10 THE EIGHT MOST IMPORTANT FACTORS IN SUCCESSFUL E-COMMERCE SITE DESIGN

| FACTOR | DESCRIPTION |
|---|---|
| Functionality | Pages that work, load quickly, and point the customer toward your product offerings |
| Informational | Links that customers can easily find to discover more about you and your products |
| Ease of use | Simple fool-proof navigation |
| Redundant navigation | Alternative navigation to the same content |
| Ease of purchase | One or two clicks to purchase |
| Multi-browser functionality | Site works with the most popular browsers |
| Simple graphics | Avoids distracting, obnoxious graphics and sounds that the user cannot control |
| Legible text | Avoids backgrounds that distort text or make it illegible |

**Table 4.10, Page 241**

# Tools for Interactivity and Active Content

- Web 2.0 design elements: Widgets, Mashups
- CGI (Common Gateway Interface)
- ASP (Active Server Pages)
- Java, JSP, and Javascript
- ActiveX and VBScript
- Coldfusion

*Insight on Technology*

# Pumping Up the Customer Experience Using AJAX and Flash

## Class Discussion

- What is AJAX?  How does it work?
- How does AJAX improve on client/server interactivity?
- How does Google Maps use AJAX?
- What are some alternative technologies to achieve the same results as AJAX?

# Personalization Tools

- Personalization
  - Ability to treat people based on personal qualities and prior history with site
- Customization
  - Ability to change the product to better fit the needs of the customer
- Tools to achieve:
  - Cookies

# The Information Policy Set

- Privacy policy
  - Set of public statements declaring how site will treat customers' personal information that is gathered by site
- Accessibility rules
  - Set of design objectives that ensure disabled users can affectively access site